# UMMAY MAIMONA CHAMAN

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LinkedIn GitHub ORCID Google Scholar Portfolio

## RESEARCH INTERESTS

Trustworthy and Human-Centered AI, with a focus on Medical AI, Computer Vision, Multimodal Deep Learning, Medical Image Analysis, Federated Learning, Wearable Health and Physiological Signal Analysis, Explainable AI (XAI), Agentic AI, LLM-based Systems, Retrieval-Augmented Generation (RAG), and Agentic AI for Physics-Based Problems.

## EDUCATION

**BRAC University** **Dhaka, Bangladesh**
*Bachelor of Science in Computer Science and Engineering* *2022 – Sep 2026*

- **CGPA:** 3.23 / 4.00
- **Relevant Coursework:** Machine Learning, Deep Learning, Neural Network, Computer Vision, Data Science, Algorithms and Data Structures, Microprocessors, Robotics, Web Development
- **Undergraduate Thesis:** *A twin-aware multimodal deep learning framework with optimized late fusion for early prediction of adolescent anxiety disorder. Under review at PLOS ONE.* Available as a preprint: medRxiv
- **Supervisor:** Dr. Md. Golam Rabiul Alam **Co-Supervisor:** Mr. Rafeed Rahman

**Holy Cross College** **Dhaka, Bangladesh**
*Higher Secondary Certificate (HSC), Science* *2019 – 2021*

- **GPA:** 5.00 / 5.00

**Monipur High School and College** **Dhaka, Bangladesh**
*Secondary School Certificate (SSC), Science* *2017 – 2019*

- **GPA:** 5.00 / 5.00

## PROFESSIONAL EXPERIENCE

**Part-Time Research Assistant | BRAC University** **Nov 2026 – Present**
*RSGI-Funded Research Project: Multimodal AI for Attention Monitoring and Stress Detection* *Dhaka, Bangladesh*

- Contribute to an RSGI-funded project on multimodal attention monitoring and stress detection using wearable sensors and computer vision in academic settings.
- Perform dataset curation, preprocessing, annotation, and quality assessment to support multimodal AI model development and evaluation.
- Collaborate with the PI, Mohammad Fahim, and Co-PIs, Md. Saif Kabir and Dr. Md. Golam Rabiul Alam, on research design, model development, and experimentation.

**Research Analyst Intern | Moncho.ai** **Sep 2026 – Present**
*AI-Assisted Market Intelligence and Agentic Research* *Dhaka, Bangladesh*

- Conduct market intelligence research covering market landscapes, competitive ecosystems, market sizing, and value-chain analysis by synthesizing information from diverse public and proprietary data sources.
- Develop AI-assisted research workflows leveraging large language models (LLMs), retrieval-augmented generation (RAG), and agentic systems for information retrieval, data extraction, synthesis, and evidence-backed analysis.
- Build and evaluate structured agentic workflows using Mastra and TypeScript, integrating tools, retrieval pipelines, and multi-step reasoning while evaluating outputs for factual consistency, source reliability, and evidence grounding.

## RESEARCH EXPERIENCE

**Undergraduate Thesis Researcher | GitHub** **Jan 2025 – June 2026**
*Supervised by Dr. Md. Golam Rabiul Alam, Professor* *BRAC University, Dhaka, Bangladesh*

- Developed a twin-aware multimodal deep learning framework for early prediction of adolescent anxiety disorder using the Queensland Twin Adolescent Brain (QTAB) dataset, integrating MRI neuroimaging (3D CNN), behavioral questionnaires, and phenotypic data through optimized weighted late fusion.
- Designed family-aware evaluation protocols to prevent data leakage in twin cohorts, achieving an ROC-AUC of 0.8935 with 85.7% sensitivity and 87.3% specificity, outperforming the strongest unimodal baseline by 11 percentage points.

**Undergraduate Research Assistant | Vision Lab** **Jan 2026 – Present**

*Research Group under Dr. Aniqua Nusrat Zereen, Assistant Professor* *BRAC University, Dhaka, Bangladesh*

- Conducting research on the RSGI-funded project *HARE: A Large-Scale Indoor RGB Video Dataset for Robust Fall Detection across Lighting Conditions*, focusing on computer vision and AI for healthcare applications.
- Contributing to dataset development and curation, experimental design, model evaluation, and manuscript preparation. Currently working on Agentic AI and Multi-Agent Systems for fall detection.

**Researcher – Agentic AI for Physics-Based Problems** **Nov 2026 – Present**

*Research Group under Dr. Aniruddha Bora, Assistant Professor* *SPARKS Lab, Texas State University, USA*

- Exploring Agentic AI approaches for solving physics-based and scientific computing problems.
- Investigating scientific machine learning, physics-informed AI, and interpretable computational modeling.
- Currently developing an evidence-grounded agentic AI framework for scientific paper review, focusing on automated critique verification, evidence retrieval, and reliable feedback generation.

**Researcher – Deep Learning for Physiological Signals and Wearable Health** **Nov 2026 – Present**

*Research Group under Dr. Ahsan Habib, Lecturer* *Deakin University, Australia*

- Conducting research on deep learning for physiological signals, wearable technologies, and interpretable medical AI.
- Investigating RWKV-based sequence modeling for ECG signal analysis, with a focus on identifying sleep-related disorders and exploring efficient long-range physiological signal modeling.

**Undergraduate Researcher** **Jan 2025 – Present**

*Research Group under Md. Saif Kabir, Senior Lecturer* *BRAC University, Dhaka, Bangladesh*

- Build and evaluate structured agent workflows using Mastra and TypeScript, integrating tools, retrieval, and multi-step reasoning while assessing AI-generated outputs for factual consistency, source reliability, and evidence grounding.
- Contributed to literature review, experimental design, model implementation, performance evaluation, and scientific manuscript preparation, resulting in one Q1 journal acceptance and one international conference paper acceptance.

## PUBLICATIONS & MANUSCRIPTS

### Journal Publications

- Noor, A., **Chaman, U. M.**, & Kabir, M. S. Federated Learning in Healthcare: A Comprehensive Survey on Privacy, Scalability, and Clinical Applications. *ICT Express (Q1 Journal).* [DOI] **Accepted.**

### Conference Publications

- Kabir, M. S., Noor, A., **Chaman, U. M.**, & Mehreen, N. Robust Multi-Backbone Hybrid Fusion for Chest X-Ray Pneumonia Detection. *International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN 2026).* [IEEE Xplore] **Accepted.**
- Antara, A., **Chaman, U. M.**, & Noor, A. SAFE-TWIN: Causal and Uncertainty-Aware Joint Charging Recommendation and Anomaly Diagnosis for Electric Vehicles. *Accepted at the IEEE International Women in Engineering Conference on Electrical and Computer Engineering (WIECON-ECE 2026).*

### Manuscripts Under Review

- Taosif, M., **Chaman, U. M.**, Prova, N. A., Taher, S. M., Alam, M. G. R., & Rahman, R. A twin-aware multimodal deep learning framework with optimized late fusion for early prediction of adolescent anxiety disorder. *medRxiv.* [Preprint] **Under Review at PLOS ONE, Q1.**
- Rabbi, R., Siraj, F. M., **Chaman, U. M.**, Noor, A., Ahmed, Z., & Zereen, A. N. HARE: A Large-Scale Indoor RGB Video Dataset for Robust Fall Detection across Lighting Conditions. **Image and Vision Computing (Q1 Journal)**
- **Chaman, U. M.**, Noor, A., Onon, T. T., Polock, A. A., Toasif, M., & Kabir, M. S. FedCXR-VQA: Multimodal FL with Vision-Language Fusion and PatchCore for Privacy-Preserving Chest X-Ray Pneumonia Diagnosis. *Under Review at the 2026 International Conference on Electrical & Computer Engineering* ***ICECE 2026***

### Manuscripts in Preparation

- **Chaman, U. M.**, & Bora, A. From Critique to Verification: Evidence-Grounded Agentic Review of Scientific Papers. *Manuscript in Preparation for Submission to a Q1 Journal.*

## TECHNICAL PROJECTS

### AI, Machine Learning and Data Science

**FaceGuard: Deepfake Detection Framework** | *PyTorch, MesoNet, Capsule Networks* | GitHub

- Designed a deepfake detection framework combining MesoNet and Capsule Networks for robust detection of manipulated facial media.

**ApneaSense: Explainable Sleep Apnea Risk Prediction** | *Python, XGBoost, SHAP* | GitHub

- Developed an ensemble learning framework for early sleep apnea risk stratification with SHAP-based explainability for interpretable clinical predictions.

**PatchCore Skin Cancer Classifier** | *PatchCore, PyTorch, Scikit-Learn, OpenCV* | GitHub | Report

- Implemented a PatchCore-based anomaly detection framework for automated skin cancer screening from dermatological images and benchmarked its performance under limited labeled-data settings.

**Multilingual RAG Chatbot: Document Intelligence System** | *FAISS, Tesseract OCR, FastAPI* | GitHub | Demo

- Built a multilingual Bengali/Banglish and English RAG system for contextual question answering over PDFs and images using OCR, semantic retrieval, and vector search.
- Developed a FastAPI backend and Streamlit frontend and deployed the system using Docker on Hugging Face Spaces.

**Japan Earthquake Occurrence Forecasting** | *Python, Pandas, NumPy, Scikit-Learn, XGBoost, SHAP* | GitHub

- Developed a spatiotemporal earthquake forecasting pipeline using 21,954 seismic events, applying feature engineering and chronological validation to predict $M \geq 5.0$ events, achieving 0.449 PR-AUC with Logistic Regression.

**Unsupervised Neural Network Architectures for Pattern Discovery** | *Python, SOM, Autoencoders* | GitHub

- Implemented Self-Organizing Maps and Autoencoders for representation learning, clustering, and hidden pattern discovery in structured datasets.

### Software Engineering / Web Development

**DataFlow Studio: Visual ETL Platform** | *Django, React, TypeScript, Pandas, REST API* | GitHub | Live Demo

- Developed a full-stack visual ETL platform with Django and React + TypeScript, enabling drag-and-drop construction of data transformation workflows.

**SkillPocket: Skill Exchange Platform** | *MERN Stack, React, Node.js, MongoDB, REST API* | GitHub | Live Demo

- Architected a full-stack skill exchange platform with React, Node.js/Express, and MongoDB, implementing authentication, scheduling, feedback, and RESTful APIs.

**AgriNetwork: Full-Stack Agricultural Platform** | *MERN stack, WebSockets* | GitHub | Live Demo

- Implemented JWT authentication, role-based access control, SSLCommerz payment integration, real-time communication using WebSockets, and RESTful APIs with production deployment on Vercel.

**RxCare: Pharmacy Management System** | *PHP, MySQL, HTML/CSS/JavaScript* | GitHub

- Designed a pharmacy management system with role-based access control, inventory monitoring, stock alerts, and a MySQL backend.

## TECHNICAL SKILLS

**Programming & Data Science:** Python (Primary), R, C/C++, SQL; Pandas, NumPy, SciPy
**Machine Learning & Deep Learning:** PyTorch, TensorFlow, Scikit-learn; Supervised & Unsupervised Learning, Model Evaluation, Feature Engineering, Hyperparameter Tuning
**Deep Learning & Computer Vision:** CNNs, RNNs, Transfer Learning, Multimodal Learning, Anomaly Detection (PatchCore), GradCAM, Medical Image Analysis
**Trustworthy AI & Healthcare Research:** Federated Learning, Explainable AI (XAI), Biomedical Data Analysis, Clinical Data Modeling, Multimodal Fusion
**LLM & Agentic AI:** Retrieval-Augmented Generation (RAG), LLM Evaluation, Agentic AI Workflows, Prompt Engineering, Information Retrieval, Evidence-Grounded Generation
**Data Handling & Visualization:** Matplotlib, Seaborn, Plotly; Statistical Analysis, Experimental Result Interpretation
**Tools & Research Workflow:** Git, Linux/Unix, Jupyter Notebook, Google Colab, LaTeX; Literature Review
**Backend & Deployment:** FastAPI, Flask, REST APIs, Docker; Streamlit, Hugging Face Spaces

## HONORS & AWARDS

- **Bronze Award** — Duke of Edinburgh International Award Program (2023).
- **Bronze Award** — The Queen's Commonwealth Essay Competition, Senior Category (2021).
- **Pre-finalist** — International Astronomy and Astrophysics Competition (IAAC) (2022).
- **Pre-finalist** — International Youth Math Challenge (IYMC) (2021).
- **Silver Badge** — HackerRank

## PROFESSIONAL TRAINING & CERTIFICATIONS

- **DataCamp** – Data Science with Python: Introduction to Data Science in Python; Exploratory Data Analysis in Python; Introduction to Statistics in Python; Supervised Learning with scikit-learn; Feature Engineering for Machine Learning in Python; Time Series Analysis in Python; Introduction to PySpark
- **Machine Learning Fundamentals** — Simplilearn (Advanced ML code and theory)
- **Introduction to Artificial Intelligence** — University of Helsinki
- **Data Science Fundamentals by Python** — DataCamp (Data Science code and theory)
- **Python Programming** — Kaggle, HackerRank (Advanced certifications)
- **JavaScript Foundations** — Cisco Networking Academy (Web development certification)
- **Digital Marketing Essentials** — Google Digital Garage

## EXTRACURRICULAR ACTIVITIES & LEADERSHIP

**Former Director** **Sep 2023 – Feb 2026**

*BRAC University Research for Development Club (BUReD)* *Dhaka, Bangladesh*

- Led research initiatives and mentored undergraduate students in research methodology, academic writing, and publication strategies.
- Organized technical workshops, research seminars, and competitions to promote research culture and interdisciplinary collaboration.
- Contributed to *BUReD Reflection Magazine* (Vol. 2 & 3) and authored two accepted articles on AI-driven image editing security and plastic waste management in Bangladesh.

**Former Undergraduate Researcher** **Jan 2024 – 2025**

*Biomedical Science and Engineering Research Center (BIOSE), BRAC University* *Dhaka, Bangladesh*

- Conducted AI-driven biomedical research in bioinformatics, computational biology, medical imaging, and disease diagnosis under faculty supervision.
- Developed and curated a dataset for fall detection applications.
- Contributed to literature reviews, manuscript preparation, and citation management.

**Former Apprentice** **2022 – 2023**

*BRAC University Robotics Club (ROBU)* *Dhaka, Bangladesh*

- Completed foundational robotics training covering sensors, microcontrollers, and basic robotics concepts through hands-on engineering activities.